Supply-Side Teacher Forecasting Framework

Primary and Secondary Education in Tanzania

Prepared for:

Ministry of Education, Science and Technology (MOEST)
Prime Minister's Office - Regional Administration and Local Government (PMO-RALG)

A technical framework for workforce planning, budgeting, and teacher deployment.

March 2026

Strategic Context

- Tanzania has expanded access to basic and secondary education through fee-free education, curriculum reforms, and school expansion.
- These gains have increased **demand for teachers**, especially across growing enrolment areas.
- Persistent constraints include fiscal pressures, uneven deployment, limited training capacity in some subjects, and attrition.
- Acute shortages remain visible in **STEM**, languages, special needs education, and remote localities.

Why forecasting?

A regular forecasting system enables MOEST and PO-RALG to shift from reactive staffing decisions to **proactive workforce planning**.

Purpose and Scope

Purpose

To provide a **clear, repeatable, and transparent** process for forecasting the supply of primary and secondary teachers in Tanzania.

What the framework supports

- Recruitment planning
- Budgeting and wage-bill discussions
- Teacher deployment and post allocation
- Scholarship and training decisions

What it covers

- Current stock of teachers
- New graduates and re-entrants
- Retirements, mortality, and attrition
- Disaggregation by level, subject, gender, and region

Institutional Arrangements and Governance

- **MOEST**: policy direction, national standards, technical oversight, and use of forecasts for teacher education and sector planning.
- **PO-RALG**: deployment, LGA coordination, stock and attrition data, and local interpretation of forecast results.
- **Teacher education institutions, TCU, NACTVET**: intake, completion, and subject specialization data.
- **Ministry of Finance, NBS, development partners**: wage-bill implications, demographic projections, and technical support.

Recommended coordination mechanism

A joint **Teacher Workforce Planning Technical Working Group** should review data, validate assumptions, agree scenarios, and sign off forecast outputs.

Forecasting Methodology

Core approach

- Stock-and-flow model as the main forecasting backbone
- Trend-based and ratio-based projections where detailed data are incomplete
- Scenario analysis to test plausible alternative futures
- Structured expert judgment to refine assumptions

Why this approach?

It is **transparent**, uses relatively low data requirements, supports **data-driven decisions**, and can be institutionalized while maintaining attention to **equity, gender, and inclusion**.

Core Stock-Flow Identity

Dynamic accounting relationship

$$\text{Stock}_{t+1} = \text{Stock}_t + \text{Inflows}_t - \text{Outflows}_t$$

Stock

- Teachers in post
- Actively available to teach

Inflows

- New graduates recruited
- Re-entries
- Transfers in

Outflows

- Retirements
- Mortality
- Attrition and exits

Interpretation

The model updates the teacher workforce **year by year**, making it suitable for annual planning and budgeting cycles.

Inflows and Outflows

Inflows

$$H_t = \text{Intake}_{t-k} \cdot c \cdot \delta$$

- Training intake drives future graduate supply
- c: completion rate
- δ: recruitment or absorption rate
- Re-entries add a smaller but relevant source of supply

Outflows

$$\text{Outflow}_t = \text{Ret}_t + \text{Mort}_t + \text{Attr}_t$$

- Retirement is relatively predictable
- Mortality is usually small but non-negligible
- Attrition is the **most policy-sensitive** component

Scenarios and Validation

Scenario design

- Baseline: continuation of current trends
- Optimistic: stronger training capacity and retention
- Pessimistic: fiscal tightening or higher attrition

Validation

- Back-testing with historical data
- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Mean Absolute Percentage Error (MAPE)

Planning value

Scenario analysis supports **risk-aware teacher workforce planning** rather than dependence on a single point forecast.

Implementation Roadmap

1. **Phase 1: Pilot and prepare**
   - Map data sources, assess quality, integrate historical data
   - Pilot a basic stock-flow model for selected regions or subjects
2. **Phase 2: Scale up**
   - Extend to all regions and subject groups
   - Align forecasting cycle with budgeting and recruitment decisions
3. **Phase 3: Consolidate and enhance**
   - Introduce more advanced statistical modules where data permit
   - Integrate supply and demand forecasting for fuller workforce planning

Conclusion and Policy Message

Core message

A practical supply-side forecasting framework allows Tanzania's education system to move from **reactive staffing responses** toward **strategic teacher workforce management**.

- It strengthens recruitment planning, deployment, and budgeting.
- It makes assumptions explicit and testable.
- It improves attention to equity, gender balance, and underserved areas.
- It can begin with existing tools and evolve as data quality improves.

Every learner should have access to a qualified teacher.